2 For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

Code:

```
import pandas as pd

data = pd.read_csv("data.csv")
X = data.iloc[:, :-1].values
Y = data.iloc[:, -1].values

S = ['0'] * X.shape[1]
G = [['?'] * X.shape[1]]

for x, y in zip(X, Y):
    if y == 'Yes':
        for i in range(len(S)):
            if S[i] == '0':
                S[i] = x[i]
            elif S[i] != x[i]:
                S[i] = '?'
        G = [g for g in G if all(
            g[i] == '?' or g[i] == S[i] for i in range(len(S))
        )]
    else:
        new_G = []
        for g in G:
            for i in range(len(g)):
                if g[i] == '?':
                    for value in data.iloc[:, i].unique():
                        if value != x[i] and value == S[i]:
                            new_g = g.copy()
                            new_g[i] = value
                            new_G.append(new_g)
        G = new_G

print("Specific Boundary S:", S)
print("General Boundary G:", G)
```

Output:

```
Specific Boundary S: ['Sunny', 'Warm', '?']
General Boundary G: [['Sunny', '?', '?'], ['?', 'Warm', '?']]
```

